Taylor Wang
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Education

Columbia University, New York, NY | B.S. Electrical Engineering | GPA: 3.87/4.00 | Graduation: May 2027

Awards

The Gates Scholarship (2023), Oxbridge Academic Programs Exceptional Merit Scholarship (2022), CIEE Global Navigator Scholarship (2021)

Portfolio Link: https://taylwang.github.io/portfolio/

Work Experience

NASA Jet Propulsion Laboratory, La Cañada, CA | Microwave Systems Intern | Jun 2025 - Aug 2025 & Jun 2026 - Aug 2026

- Estimated affine transformation parameters using an iterative least-squares optimization algorithm for satellite image registration.
- Implemented a Bayesian super-resolution algorithm using Langevin dynamics and Field-of-Experts priors to reconstruct high-resolution Mars orbital imagery.
- Mapped signal transformations across a radiometer using Fourier analysis, improving debugging efficiency during system validation.
- Diagnosed a signal inversion in the radiometer responsible for erroneous correlation measurements, enabling accurate downstream analysis and interpretation.

Columbia Wireless and Mobile Networking Lab, New York, NY | Research Assistant | Oct 2025 - Jan 2026

- Recreated New York State's proprietary neural network for converting radiometer data into water vapor concentration, enabling analysis of model robustness under radio frequency interference.
- Developed a Python pipeline to detect and classify 5G radio frequency interference in radiometer measurements.

UW-Madison Woods Group, Madison, WI | Computational Assistant | May 2024 - Sep 2024

- Solved the 2D Schrödinger equation numerically using finite-difference methods and sparse matrix techniques to obtain eigenvalues and eigenvectors of quantum states.
- Separated contributions from distinct quantum valleys by constructing a localized basis, enabling the calculation of intra-valley and inter-valley coupling.

Mad City Labs, Madison, WI | Programmer | May 2023 - Aug 2023

- Developed a Java-based graphical user interface for controlling company microscope systems.
- Integrated open-source microscope control modules into the user interface, eliminating the need to develop communication protocols for four hardware devices.

Personal Project

Automated Left Ventricle Identification in Echocardiograms | Sep 2025 - Dec 2025

- Developed an automated image-processing pipeline in Python to identify the left ventricle in echocardiogram images.
- Designed an image preprocessing pipeline using median and anisotropic Gaussian filtering to enhance ventricle boundaries while suppressing imaging artifacts.
- Applied watershed segmentation to generate candidate left ventricle regions.
- Devised a weighted scoring algorithm using centroid location, area, and shape metrics to identify the correct left ventricle from segmented candidates.

Skills

Core Competencies: Signal Processing, Image Processing, Computer Vision, Numerical Optimization
Mathematics: Linear Algebra, Probability, Optimization, Fourier Analysis.
Programming: Python, Java, MATLAB.